Geometry

Labix

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Contents

1 Geometry of Metric Spaces

1.1 Distance Preserving Maps

Definition 1.1.1 (Distance Preserving Maps)
Let (X, d_X) and (Y, d_Y) be metric spaces. Let $f : X \to Y$ be a function. We say that f is distance preserving if for any $p, q \in X$, we have

$$d_Y(f(p), f(q)) = d_X(p, q)$$

Lemma 1.1.2
Let (X, d_1) and (Y, d_2) be metric spaces. Let $f : X \to Y$ be a distance preserving map. Then f is continuous.

Proof Let $p \in X$. Let $\varepsilon > 0$. Choose $\delta = \varepsilon$, then for all $x \in X$ such that $0 < d_X(x, p) < \delta$, we have $d_Y(f(x), f(p)) = d_X(x, p) < \delta = \varepsilon$ so that f is continuous. ■

Definition 1.1.3 (Isometries) Let (X, d_1) and (X, d_2) be metric spaces. Let $f : X \to Y$ be a distance preserving map. We say that f is an isometry if there exists a distance preserving map $g : Y \to X$ such that

$$g \circ f = \mathrm{id}_X \quad \text{and} \quad f \circ g = \mathrm{id}_Y$$

It is also common to use the term isometry to mean a distance preserving function and refer to our version of isometry as bijective isometries.

Lemma 1.1.4 Let (X, d_1) and (X, d_2) be metric spaces. Let $f : X \to Y$ be a distance preserving map. Then f is an isometry if and only if f is bijective.

Proof
Clearly, if f is an isometry then f has an inverse function. Conversely, suppose that f is bijective. Then we have

$$d_X(f^{-1}(p), f^{-1}(q)) = d_Y(f(f^{-1}(p)), f(f^{-1}(q))) = d_Y(p, q)$$

for any $p, q \in Y$ since f is distance preserving. Hence f^{-1} is distance preserving. ■

Lemma 1.1.5
Let (X, d) be a metric space. Then

$$\text{Isom}(X, d)$$

is a group.

Proof
We have seen from Metric Spaces that inverses and compositions are isometries. Moreover, the identity is clearly an isometry. Hence $\text{Isom}(X, d)$ is a group. ■

1.2 Isometric Isomorphisms

Definition 1.2.1 (Isometric Isomorphisms)
Let V, W be normed vector spaces. Let $T : V \to W$ be a function. We say that T is an isometric isomorphism if T is an isometry and a linear isomorphism.

1.3 Geodesics

Definition 1.3.1 (Euclidean Metric)
Let $x, y \in \mathbb{R}^n$. The Euclidean Metric is defined to be

$$d(x, y) = \|x - y\|$$

Clearly the Euclidean metric indeed defines a metric on $\mathbb{R}^n$. Any subspaces of $\mathbb{R}^n$ inherits the subspace metric of $\mathbb{R}^n$. In particular, $I = [0, 1]$ is given a metric.

Definition 1.3.2 (Geodesics) Let (X, d) be a metric space. A geodesic on X is a distance preserving function $\gamma : I \subseteq \mathbb{R} \to X$.

Definition 1.3.3 (Geodesic Metric Spaces) Let (X, d) be a metric space. We say that X is geodesic if for all $a, b \in X$, there exists a geodesic $\gamma : I \to X$ from a to b.

2 Affine Geometry

Affine geometry refers to the geometry done on spaces but ignoring the notions of coordinates and measures of angles and distance.

2.1 The Analytical Definition of Affine Space

We must first define a certain notion of space in order to discuss affine geometry. These spaces which has the barebone structure needed to talk about affine geometry is called affine spaces. We would like affine spaces to have the following properties.

- Aside from the underlying set of the space, we would like a set of vectors so that we can add a vector with a given point into a new point.
- Given any two points x, y in the space, there is exactly one vector that sends x to y.

We take the following definition for the underlying space of affine geometry.

Definition 2.1.1 (Affine Space)
An affine space consists of the following data.
- A set A.
- A vector space V.
- A free and transitive action of V on A.

The action of the vector space on the points give us addition of points and vectors, and the free and transitive action guarantees a unique vector connecting any two points.

Lemma 2.1.2
Let $(A, V, +)$ be an affine space. Let $v \in V$. Then the function $A \to A$ defined by $a \mapsto a + v$ is a bijection.

Definition 2.1.3 (Subtraction)
Let $(A, V, +)$ be an affine space. Let $a, b \in A$. Define the subtraction of a from b to be the unique vector $b - a \in V$ such that
$$a + (b - a) = b$$

Definition 2.1.4 (Affine Subspaces)
Let $(A, V, +)$ be an affine space. An affine subspace of A is a set of the form
$$a + W = \{a + w \mid w \in W\}$$
where $W \subseteq V$ is a subspace.

2.2 Affine Transformations

Definition 2.2.1 (Affine Transformations)
Let $(A, V, +)$ and $(B, W, +)$ be affine spaces. Let $f : A \to B$ be a function. We say that f is an affine transformation if the induced function $\overline{f} : V \to W$ defined by
$$q - p \mapsto f(q) - f(p)$$
is a well defined linear map for $p, q \in A$.

Definition 2.2.2 (Affine Isomorphism)
Let $(A, V, +)$ and $(B, W, +)$ be affine spaces. Let $f : A \to B$ be an affine transformation. We say that f is an affine isomorphism if $\overline{f} : V \to W$ is an isomorphism of vector spaces.

Proposition 2.2.3
Let $(A, V, +)$ be an affine space. Let $a \in A$. Then there is a unique affine isomorphism $f : A \to V$ such that $f(a) = 0_V$.

2.3 The Synthetic Approach

3 Euclidean Geometry

3.1 Lines and Incidences

Definition 3.1.1 (Euclidean Space)
A Euclidean space is an affine space (E, V) such that V is a finite dimensional inner product space over $\mathbb{R}$.

In Linear Algebra, we classified that up to isomorphism, every finite dimensional inner product space is isomorphic to $\mathbb{R}^n$ equipped with the standard dot product. Since we have also seen that every affine space has a unique isomorphism to its underlying vector space by specifying the point of origin, we lose no generality by working with $\mathbb{R}^n$ and the standard dot product. Therefore some literature define Euclidean space to be $\mathbb{R}^n$ equipped with its standard dot product.

Definition 3.1.2 (The Metric of Euclidean Spaces)
Let E be a Euclidean space. Define the metric $d : E \times E \to \mathbb{R}$ on E to be

$$d(P, Q) = \left|\overrightarrow{PQ}\right|$$

where $\overrightarrow{PQ}$ is the unique vector such that $P + \overrightarrow{PQ} = Q$.

Lemma 3.1.3
Let E be a Euclidean space. Then $d : E \times E \to \mathbb{R}$ is indeed a metric on E.

Definition 3.1.4 (Line Segments)
Let (E, d) be an Euclidean space. Let $P, Q \in E$. Define the line from P to Q to be

$$\overline{PQ} = \{R \in E \mid d(P, R) + d(R, Q) = d(P, Q)\}$$

If $E = \mathbb{R}^n$ instead, another way to write the set of points that form the line segment between two points P and Q is

$$\overline{PQ} = \{P + t(Q - P) \mid 0 \le t \le 1\}$$

Definition 3.1.5 (Lines)
Let (E, d) be an Euclidean space. Let $P, Q \in E$. Define the line from P to Q to be

$$L_{P,Q} = \left\{ R \in E \;\middle|\; \begin{matrix} d(P,R)+d(R,Q)=d(P,Q) \text{ or} \\ d(P,Q)+d(Q,R)=d(P,R) \text{ or} \\ d(P,Q)+d(P,R)=d(Q,R) \end{matrix} \right\}$$

Similarly, if $E = \mathbb{R}^n$ then there is an equality

$$L_{P,Q} = \{P + t(Q - P) \mid t \in \mathbb{R}\}$$

Definition 3.1.6 (Collinearity)
Let E be an Euclidean space. Let $P, Q, R \in E$. We say that P, Q, R are collinear if there exists a line $L \subseteq E$ such that

$$P, Q, R \in L$$

Lemma 3.1.7
Let E be an Euclidean space. Let $T : E \to E$ be an isometry. If $P, Q, R \in E$ are collinear, then $T(P), T(Q), T(R)$ are collinear.

Proof
Let $P, Q, R \in E$ be collinear. Without loss of generality suppose that $d(P, Q) + d(Q, R) = d(P, R)$. Since T is an isometry, we have

$$d(T(P), T(Q)) + d(T(Q), T(R)) = d(T(P), T(R))$$

Hence $T(R) \in L_{T(P),T(Q)}$ and so $T(P), T(Q), T(R)$ are collinear. ■

Lemma 3.1.8
Let E be an Euclidean space. Let $T : E \to E$ be an isometry. Let $L \subseteq E$ be a line. Then $T(L)$ is also a line.

Proof
I claim that $T(L) = L_{T(P),T(Q)}$ for any $P, Q \in L$. Let $W \in T(L)$. Then there exists $R \in L$ such that $W = T(R)$. Since $R \in L = L_{P,Q}$, P, Q, R are collinear. Hence $T(P), T(Q), T(R)$ are collinear. Thus $T(R) \in L_{T(P),T(Q)}$. Hence $T(L) \subseteq L_{T(P),T(Q)}$. Conversely, suppose that $W \in L_{T(P),T(Q)}$, then $T(P), T(Q), W$ are collinear. Since T is an isometry, there exists $R \in E$ such that $W = T(R)$. Moreover, T^{-1} is also an isometry hence $T(P), T(Q), T(R)$ collinear implies that P, Q, R are collinear. Hence $R \in L_{P,Q} = L$. Hence $R \in T(L)$ and so $T(L) = L_{P,Q}$. Hence $T(L)$ is a line.
■

Definition 3.1.9 (Parallel Lines)

Proposition 3.1.10 (The Parallel Postulate is True in Euclidean Geometry)

Definition 3.1.11 (Perpendicular Lines)

3.2 Angles

Recall that from real analysis, we defined the power series

$$\cos(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} \quad \text{and} \quad \sin(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}$$

We have seen that $\cos(x)$ is strictly decreasing in the interval $[0, \pi]$ and so $\cos(x) = k$ for $-1 \leq k \leq 1$ has a unique solution between 0 and π.

Definition 3.2.1 (Angles of a Euclidean Space)
Let E be a Euclidean space. Let $P, Q, R \in E$ be points. Define the angle between $\overline{PQ}$ and $\overline{QR}$ to be the unique real number $\angle PQR \in \mathbb{R}$ such that

$$\cos(\angle PQR) = \frac{\overrightarrow{QP} \cdot \overrightarrow{QR}}{\left|\overrightarrow{QP}\right|\left|\overrightarrow{QR}\right|}$$

and $0 \leq \angle PQR \leq \pi$.

Lemma 3.2.2
Let E be a Euclidean space. Let $P, Q, R \in E$ be points. If $Q \in \overline{PR}$, then $\angle PQR = \pi$.

Proof
Since $Q \in \overline{PR}$, we have that $d(P, Q) + d(Q, R) = d(P, R)$, which translates to $\left|\overrightarrow{PQ}\right| + \left|\overrightarrow{QR}\right| = \left|\overrightarrow{PR}\right|$. Hence the vectors $\overrightarrow{PQ}$ and $\overrightarrow{QR}$ are collinear in $\mathbb{R}^n$ as a Euclidean space. Thus there exists $\lambda > 0$

such that $-\lambda\overrightarrow{QP} = \overrightarrow{QR}$. Then we have

$$\begin{aligned}\cos(\angle PQR) &= \frac{\overrightarrow{QP}\cdot\overrightarrow{QR}}{\left|\overrightarrow{QP}\right|\left|\overrightarrow{QR}\right|} \\ &= \frac{-\lambda\overrightarrow{QP}\cdot\overrightarrow{QP}}{\lambda\left|\overrightarrow{QP}\right|\left|\overrightarrow{QP}\right|} \\ &= \frac{-\lambda\left|\overrightarrow{QP}\right|^2}{\lambda\left|\overrightarrow{QP}\right|^2} \\ &= -1\end{aligned}$$

Since $\cos(x)$ is bijective in $[0,\pi]$ and $\cos(\pi) = -1$, we conclude that $\angle PQR = \pi$. ■

Proposition 3.2.3 (Vertically Opposite Angles)
Let E be a Euclidean space. Let $P, Q, R, S \in E$ such that $\overline{PQ}$ meets $\overline{RS}$ at the unique point $T \in E$. Then we have $\angle PTR = \angle QTS$.

Definition 3.2.4 (Triangles)
Let E be a Euclidean space. A triangle in E consists of three distinct points $P, Q, R \in E$ together with the line segments

$$\triangle PQR = \overline{PQ}\cup\overline{PR}\cup\overline{QR}$$

Theorem 3.2.5 (Angle Sum of Triangle)
Let E be a Euclidean space. Let $\triangle P, Q, R \subseteq E$ be a triangle. Then we have

$$\angle PQR + \angle QRP + \angle RPQ = \pi$$

Definition 3.2.6 (Isosceles Triangle)
Let E be a Euclidean space. Let $\triangle PQR \subseteq E$ be a triangle. We say that $\triangle PQR$ is isosceles if $\overline{PQ} = \overline{PR}$ (up to a permutation of P, Q, R).

Proposition 3.2.7
Let E be a Euclidean space. Let $\triangle P, Q, R \subseteq E$ be a triangle. Then the following are equivalent.

- $\triangle PQR$ is an isosceles triangle with $\overline{PQ} = \overline{PR}$.
- $\angle PQR = \angle PRQ$.
- There exists an isometry $\phi : E \to E$ such that $\phi(P) = P$, $\phi(Q) = R$, $\phi(R) = Q$.

3.3 The Four Centers of a Triangle

3.4 Theorems of Circles

3.5 Classifying Isometries of Euclidean Space

Definition 3.5.1 (Affine Maps) A map $T : \mathbb{R}^n \to \mathbb{R}^m$ is affine if it is of the form

$$T(x) = Ax + b$$

for all $x \in \mathbb{R}^n$ for some linear map A and $b \in \mathbb{R}^k$.

Proposition 3.5.2 Let $n, k \in \mathbb{N}\setminus\{0\}$. Let $T : \mathbb{R}^n \to \mathbb{R}^k$ be a function. Then the following are equivalent.

- T is an affine map.

- The map $L(x) = T(x) - T(0)$ is a linear map.
- For all $\lambda \in \mathbb{R}$, $T(\lambda x + (1-\lambda)y) = \lambda T(x) + (1-\lambda)T(y)$

Proof Let $T : \mathbb{R}^n \to \mathbb{R}^k$ be a map.

- $(1) \iff (2)$: Define $L(x) = T(x) - T(0)$. Then T is affine if and only if L is linear, if and only if $L(\lambda x + \mu y) = \lambda L(x) + \mu L(y)$ if and only if $T(\lambda x + \mu y) - T(0) = \lambda(T(x) - T(0)) + \mu(T(y) - T(0))$.
- $(2) \implies (3)$: This is obtained by setting $\mu = 1 - \lambda$.
- $(3) \implies (2)$: We have

$$\begin{aligned} \lambda x + \mu y &= \frac{1}{2}(2\lambda x) + \frac{1}{2}(2\mu y) \\ T(\lambda x + \mu y) &= \frac{1}{2}T(2\lambda x) + \frac{1}{2}T(2\mu y) && \text{(By (3))} \end{aligned}$$

Also by (3), we have

$$\begin{aligned} 2\lambda x &= 2\lambda x + (1 - 2\lambda)0 \\ T(2\lambda x) &= 2\lambda T(x) + (1 - 2\lambda)T(0) \end{aligned}$$

And similarly,

$$T(2\mu y) = 2\mu T(y) + (1 - 2\mu)T(0)$$

Combining the three, we have

$$\begin{aligned} T(\lambda x + \mu y) &= \frac{1}{2}T(2\lambda x) + \frac{1}{2}T(2\mu y) \\ &= \frac{1}{2}(2\lambda T(x) + (1 - 2\lambda)T(0)) + \frac{1}{2}(2\mu T(y) + (1 - 2\mu)T(0)) \\ &= \lambda T(x) + \left(\frac{1}{2} - \lambda\right) T(0) + \mu T(y) + \left(\frac{1}{2} - \lambda\right) T(0) \\ T(\lambda x + \mu y) - T(0) &= \lambda(T(x) - T(0)) + \mu(T(y) - T(0)) \end{aligned}$$

■

Proposition 3.5.3 Let $n \in \mathbb{N} \setminus \{0\}$. Let $L : \mathbb{R}^n \to \mathbb{R}^n$ be a linear map. Let A be the matrix representing T under the standard basis. Then the following are equivalent.

- L is an isometry
- $\|L(x)\| = \|x\|$ for all $x \in \mathbb{R}^n$
- $\langle L(x), L(y)\rangle = \langle x, y\rangle$ for all $x, y \in \mathbb{R}^n$
- A is an orthogonal matrix.

Proof Let $L : \mathbb{R}^n \to \mathbb{R}^k$ be linear.

- $(1) \implies (2)$: This is trivial since isometries are distance preserving
- $(2) \implies (3)$: This is given by the polarization identity

- $(3) \implies (4)$: We have that

$$\begin{aligned}(A^TA)_{ij} &= \sum_{k=1}^{n}(A^T)_{ik}A_{kj} \\ &= \sum_{k=1}^{n}A_{ki}A_{kj} \\ &= \sum_{k=1}^{n}L(e_i)_kL(e_j)_k \\ &= \langle L(e_i), L(e_j)\rangle \\ &= \langle e_i, e_j\rangle && \text{(By (3))} \\ &= \delta_{ij}\end{aligned}$$

  Thus $A^TA = I$

- $(4) \implies (1)$: Suppose that $A^TA = I$. Then A is invertible and thus is a bijection. Thus we just need to show that $d(x,y) = d(L(x), L(y))$.

$$\begin{aligned}\|L(x)\|^2 &= \langle L(x), L(x)\rangle \\ &= (L(x))^TL(x) \\ &= x^TA^TAx \\ &= x^Tx \\ &= \langle x, x\rangle \\ &= \|x\|^2\end{aligned}$$

  Thus $d(L(x), L(y)) = \|L(x) - L(y)\| = \|L(x-y)\| = \|x-y\| = d(x,y)$.

■

Proposition 3.5.4 Let $n \in \mathbb{N} \setminus \{0\}$. Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a function. Then T is an isometry if and only if T is affine and given by $T(x) = Ax + b$, where A is orthogonal.

Proof We have shown that Euclidean Isometries are line preserving, thus by 1.3.2 they are affine and in the form $T(x) = Ax + b$. We now check that $G(x) = Ax = T(x) - b$ is an isometry. G is distance preserving since

$$\begin{aligned}d(G(x), G(y)) &= \|G(x) - G(y)\| \\ &= \|T(x) - b - T(y) + b\| \\ &= \|T(x) - T(y)\| \\ &= \|x - y\| && (T \text{ is an isometry}) \\ &= d(x,y)\end{aligned}$$

We now show that G is bijective. Since T is an isometry, T is bijective and $T(x) - b$ is also bijective with $T^{-1}(x) = x + b$ and thus G is an isometry. By the above proposition, A is orthogonal. ■

Proposition 3.5.5 Let $\phi : O(n,\mathbb{R}) \to \mathrm{Aut}(\mathbb{R}^n)$ be the function defined by $\phi(A)(x) = Ax$ for $x \in \mathbb{R}^n$. Then ϕ is a group homomorphism. Moreover, there is a group isomorphism

$$\psi : \mathrm{Isom}(\mathbb{R}^n, d) \to \mathbb{R}^n \rtimes_\phi O(n,\mathbb{R})$$

given by $T \mapsto (T(0), T - T(0))$.

3.6 Canonical Form of Orthogonal Matrices

Theorem 3.6.1 (Canonical Form of Orthogonal Matrices) Let $M \in O(n, \mathbb{R})$ be an orthogonal matrix. Then there exists an orthonormal basis of $\mathbb{R}^n$ such that M is similar to a matrix of the form

$$\begin{pmatrix} I_k & & & & \\ & -I_m & & & \\ & & B_1 & & \\ & & & \ddots & \\ & & & & B_l \end{pmatrix}$$

where

$$B_i = \begin{pmatrix} \cos(\theta_i) & -\sin(\theta_i) \\ \sin(\theta_i) & \cos(\theta_i) \end{pmatrix}$$

Proof Let $L : \mathbb{R}^n \to \mathbb{R}^n$ be an isometry. We prove the result by induction on n. Extend the domain of L such that it becomes a map from $\mathbb{C}^n \to \mathbb{C}^n$. By the fundamental theorem of algebra, $c_A(L)$ has at least one root, λ. We know that $|\lambda| = 1$. There are two cases.

- If $\lambda \in \mathbb{R}$, then $\lambda = \pm 1$. Choose an eigenvector $z = x + iy$, with $x, y \in \mathbb{R}^n$. We have that $\lambda(x + iy) = L(x + iy) = L(x) + iL(y)$, thus x, y respectively are both real eigenvectors of λ. At least one of these x, y is non zero, else $z = 0$. Now $\frac{x}{\|x\|}$ is a basis for $W = \mathbb{C} \cdot x$ and the matrix for L is
$$\begin{pmatrix} \pm 1 & 0 \\ 0 & L|_{W^\perp} \end{pmatrix}$$
- If $\lambda \in \mathbb{C} \setminus \mathbb{R}$, choose an eigenvector z. We must have $\overline{\lambda}$ is also an eigenvalue, with eigenvector $\overline{z}$. Let $W = E_\lambda \oplus E_{\overline{\lambda}}$. By the above lemma, there exists a real orthonormal basis for W. In terms of a basis for $W \oplus W^\perp$, the matrix for L is
$$\begin{pmatrix} \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix} & 0 \\ 0 & L|_{W^\perp} \end{pmatrix}$$

By the above, in both cases, $L|_{W^\perp}$ is an isometry of $W^\perp$, so we can apply the induction step to $W^\perp$. Eventually, we will have decomposed V into mutually orthogonal subspaces $W_1, \dots, W_M$, all invariant under L. Either W_i has dimension 2 with a real orthonormal basis with respect to which the matrix of $L|_W$ has the form as in the above lemma, or W_i has dimenison 1, with normalized real basis, and L acts as 1 or -1. Since the W_i are mutally orthogonal, the basis consisting of the union of all these basis is orthonormal. By reordering the subspaces appropriately we obtain the above form. ∎

I would say that the main part of this theorem is that you essentially decompose $\mathbb{R}^n$ into $W_1 \oplus \cdots \oplus W_n$, where they are mutually orthogonal. It might be the case that an isometry in $\mathbb{R}^2$ or $\mathbb{R}^3$ allows some orthogonal to rotate or reflect, leaving the rest invariant. Thus performing each rotation or reflection in $\mathbb{R}^3$ is only transforming one of $W_1, \dots, W_n$.

3.7 Isometry Decomposition

Definition 3.7.1 (Hyperplane)
Let $n \in \mathbb{N} \setminus \{0\}$. A hyperplane in $\mathbb{R}^n$ is a set of the form

$$\Pi = \{v + b \mid v \in V\}$$

for $V \subseteq \mathbb{R}^n$ a vector subspace of dimension $n - 1$.

Definition 3.7.2 (Fixed Points) Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be an isometry. Denote the set of fixed points of T to be

$$\text{Fix}(T) = \{x \in \mathbb{R}^n \mid T(x) = x\}$$

Definition 3.7.3 (Reflection on Hyperplane)
Let $\Pi \subseteq \mathbb{R}^n$ be a hyperplane. A reflection in Π is an isometry $T : \mathbb{R}^n \to \mathbb{R}^n$ such that

$$\text{Fix}(T) = \Pi$$

Proposition 3.7.4 The reflection on hyperplane exists and is unique.

Proof Let $\Pi = V + b$. Pick $v \in V^\perp$. Take a basis of $\mathbb{R}^n$ consisting of v together with a basis for V. With respect to the basis, define a linear map T with matrix

$$A = \begin{pmatrix} -1 & 0 \\ 0 & I_{n-1} \end{pmatrix}$$

Note that this fixes V and is not an identity. Let $S(x) = x - b$. I claim that $\rho = S^{-1} \circ T \circ S$ is a reflection. I prove that $\text{Fix}(\rho) = \Pi$. This means solving $x = Ax + (I - A)b$. We have that $(A - I)(x - b) = 0$. Solving this gives

$$x = \begin{pmatrix} 0 + b_1 \\ t_2 + b_2 \\ \ddots \\ t_n + b_n \end{pmatrix}$$

But then $x \in \Pi$ since x is of the form $b + v$ with $v \in V$. Thus ρ_Π exists.
Let ρ' that is not the identity mapping fixes Π. Then $R = T \circ \rho' \circ T^{-1}$ by 1.4.2 fixes V and fixes $V^\perp$. Thus $R(v) = \lambda v$ for some $\lambda \in \mathbb{R}$. Since R is an isometry, $|\lambda| = 1$. Since R is not the identity, $\lambda = -1$ and thus R has matrix A. Thus $\rho = \rho'$ and we have proved uniqueness. ■

Lemma 3.7.5 Let $p \in \mathbb{R}^n$. Let $V \subseteq \mathbb{R}^n$ and $v \perp V$. Let ρ_Π be the reflection on a hyperplane $\Pi = V + b$. Then

$$\rho_\Pi(p) = p - 2\langle p - b, v\rangle v$$

Proof Let d be the shortest Euclidean distance between P and the hyperplane. To obtain the reflection of P along the hyperplane, we calculate the vector that starts at P with magnitude d and direction towards the hyperplane, and perpendicular to it, then add it to P two times to obtain it.

Let the point where the tail of v is on Π be Q. Note that $QP = P - b$. Since $\|v\| = 1$, we have that the magnitude of the projection of QP on to v is given by $\langle P - b, v\rangle$. Thus using our method, we obtain that $\rho_\Pi(P) = P - 2\langle P - b, v\rangle v$. ■

Lemma 3.7.6 Let $p, q \in \mathbb{R}^n$ be distinct. Then there exists a reflection ρ with $\rho(p) = q$.

Proof Let $v = \frac{p-q}{\|p-q\|}$. Let $b = \frac{p+q}{2}$. I claim that $\Pi = \mathbb{R}v^\perp + b$ admits a reflection such that $\rho_\Pi(p) = q$.

$$\begin{aligned} \rho_\Pi(p) &= p - 2\left\langle p - \frac{p+q}{2}, \frac{p-q}{\|p-q\|} \right\rangle \frac{p-q}{\|p-q\|} \\ &= p - \langle p - q, p - q\rangle \frac{p-q}{\|p-q\|^2} \\ &= q \end{aligned}$$

Thus we are done. ■

Theorem 3.7.7 Let $T \in \text{Isom}(\mathbb{R}^n, d)$ be an isometry of $\mathbb{R}^n$. Then T is the composition of at most $n+1$ reflections.

Proof Let T be an isometry. Let $P = T(0)$ and $Q = 0$. Then by the above, there exists a reflection R such that $R(T(0)) = 0$. Thus $L = R \circ T$ is a linear isometry. Choose an orthonormal basis $v_1, \dots, v_n$ so that by the normal form theorem, the matrix M has the following form

$$M = \begin{pmatrix} I_k & & & & \\ & -I_m & & & \\ & & B_1 & & \\ & & & \ddots & \\ & & & & B_l \end{pmatrix}$$

Denote J_i to be the $n \times n$ identity matrix, except -1 is at the (i,i)th spot. Note that this is the matrix of a reflection in $(\mathbb{R}v_i)^\perp$. Let B_θ be the matrix of rotation of degree θ. Then it can be decomposed into

$$B_i = A_i \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} \cos(\theta_i) & \sin(\theta_i) \\ \sin(\theta_i) & -\cos(\theta_i) \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

The matrix A_θ has eigenvalues 1 and -1, with corresponding eigenvectors $\left(\cos\left(\frac{\theta_i}{2}\right), \sin\left(\frac{\theta_i}{2}\right)\right)$ and $\left(\sin\left(\frac{\theta_i}{2}\right), -\cos\left(\frac{\theta_i}{2}\right)\right)$ respectively.

Define $C_i = J_{k+m+2i}$ and

$$D_i = \begin{pmatrix} I_{k+m+2i-2} & 0 & 0 \\ 0 & B_i & 0 \\ 0 & 0 & I_{n-k-m-2i} \end{pmatrix}$$

and

$$E_i = \begin{pmatrix} I_{k+m+2i-2} & 0 & 0 \\ 0 & A_i & 0 \\ 0 & 0 & I_{n-k-m-2i} \end{pmatrix}$$

Note that E_i is a reflection in the plane orthogonal to $\sin\left(\frac{\theta_i}{2}\right) v_{k+m+2i-1} - \cos\left(\frac{\theta_i}{2}\right) v_{k+m+2i}$. Then we have

$$M = J_{k+1} \circ \cdots \circ J_{k+m} \circ D_1 \circ \cdots \circ D_l$$

and thus

$$M = J_{k+1} \circ \cdots \circ J_{k+m} \circ E_1 \circ C_1 \circ \cdots \circ E_l \circ C_l$$

which is the product of at most n reflections. Since $T = R^{-1} \circ L$, T is the product of at most $n+1$ reflections. ■

Corollary 3.7.8
Let $n \in \mathbb{N} \setminus \{0\}$. Then we have

$$\text{Isom}(\mathbb{R}^n, d) = \langle T : \mathbb{R}^n \to \mathbb{R}^n \mid T \text{ is a reflection } \rangle$$

Proof
The above theorem implies that every isometry is the composition of finitely many reflections. ■

4 Spherical Geometry

4.1 The n-Sphere

Definition 4.1.1 (n-Dimensional Sphere) Let $r \geq 0$. Define the n dimensional sphere of radius r to be
$$S^n(r) = \{(x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} : \|x\| = r\}$$

Definition 4.1.2 (Great Circle) A great circle is the intersection of $S^n(r)$ with a two dimensional vector subspace of $\mathbb{R}^{n+1}$

Definition 4.1.3 (Antipodal Points) Let $P, Q \in S^n(r)$. We say that P and Q are antipodal if $Q = -P$.

Lemma 4.1.4 If $P, Q \in S^n(r)$ are not antipodal, then there exists a unique great circle containing P and Q.

Definition 4.1.5 (Collinearity) Three distinct points on S^n are said to be collinear if there exists a great circle such that the three points lie on it.

Definition 4.1.6 (Spherical Triangle) A triangle on $S^n(r)$ is three distinct points that are joined together by three great circles.

4.2 Spherical Metric

Definition 4.2.1 (Spherical Metric) Define the Spherical Metric by
$$d_S(P, Q) = r \cos^{-1}\left(\frac{\langle P, Q\rangle}{r^2}\right)$$
where we take the domain of $\cos^{-1}$ to be $[0, \pi]$.

Proposition 4.2.2 Let $\triangle PQR$ be a spherical triangle of S^n. Then $\angle QOR = d(Q, R)$ and vice versa for the other angles.

Proof By definition of a radian, the arc QR is equal to the angle $\angle QOR$. ∎

Definition 4.2.3 (Spherical Angle) Suppose that two great circles on S^n intersect at one of the points P. Define the spherical angle P to be the smaller angle made on S^n between the two great circles. We write it as $\angle_S$.

Proposition 4.2.4 Let PQR be a spherical triangle of S^2. Suppose that $\alpha = \angle_S QPR$, $\beta = \angle_S PRQ$ and $\gamma = \angle_S PQR$. Let $a = \angle QOR = d_{S^2}(Q, R)$. Then
$$\cos(\alpha) = \cos(\beta)\cos(\gamma) + \sin(\beta)\sin(\gamma)\cos(a)$$

Proof Note that isometries on $\mathbb{R}^3$ preserves the inner product, since spherical distance is defined by the inner product, the spherical distance is preserves. Thus we may perform an isometry such that $P = (0, 0, 1)$. We can then perform a rotation such that Q maps to a point on the xz plane. We can then write $Q = (\sin(\beta), 0, \cos(\beta))$ and $R = (\sin(\gamma)\cos(\alpha), \sin(\gamma)\sin(\alpha), \cos(\gamma))$. Calculating this gives our required formula. ∎

Proposition 4.2.5 (Triangle Inequality) Let PQR be a spherical triangle of S^n whose sides are shorter arcs given by $\alpha, \beta, \gamma \leq \pi$. Then

$$\alpha \leq \beta + \gamma$$

with equality if and only if PQR is collinear.

Proof Notice that by definition, $\alpha, \beta, \gamma \in [0, \pi]$ thus $\sin(\beta)$ and $\sin(\gamma)$ is greater than or equal to 0. Since $\cos(a) \geq -1$, we must have from the above proposition,

$$\begin{aligned} \cos(\alpha) &\geq \cos(\beta)\cos(\gamma) - \sin(\beta)\sin(\gamma) \\ &= \cos(\beta + \gamma) \end{aligned}$$

We split it into two cases. If $\beta + \gamma \leq \pi$, then $\cos(x)$ being decreasing function on $[0, \pi]$ implies $\alpha \leq \beta + \gamma$ and we are done. Equality is given as long as $\cos(a) = -1$, which means that $a = \pi$, which forces $\beta + \gamma = \pi$ from the inequality.
Now if $\beta + \gamma > \pi$, then the triangle inequality is trivial since $\alpha \leq \pi < \beta + \gamma$. Now equality occurs if and only if $a = \pi$. Then this is true if and only if $\cos(\alpha) = \cos(\beta + \gamma)$ which is true if and only if $\alpha + \beta + \gamma = 2\pi$ and we are done. ■

Proposition 4.2.6 The spherical metric is indeed a metric on S^n. And thus S^n is a metric space.

Proof ■

4.3 Isometries in S^n

Proposition 4.3.1 An isometry on S^n preserves antipodal points and great circles.

Proposition 4.3.2 Every isometry in S^n can be extended to a Euclidean Isometry of $\mathbb{R}^{n+1}$. Every Euclidean isometry $T : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ can be restricted to an spherical isometry $T|_{S^n} : S^n \to S^n$.

Proposition 4.3.3 Every isometry $T : S^n \to S^n$ is of the form $T(x) = Ax$ for all $x \in S^n$, where $A \in O(n+1)$.

Corollary 4.3.4 There exists a group isomorphism

$$\text{Isom}(S^n, d_{S^n}) \cong O(n+1)$$

4.4 Geometry of the 2-Sphere

Proposition 4.4.1 Let C, D be two distinct great circles, then their intersection is a pair of antipodal points.

Theorem 4.4.2 (Girad's Theorem) Let $\triangle PQR$ be a spherical triangle on S^2. With P, Q, R distinct points and such that the interior of $\triangle PQR$ does not intersect the great circles which contain the sides of $\triangle PQR$. Then the area of the triangle is equal to $\angle PQR + \angle PRQ + \angle QPR - \pi$.

The reason that we have this much constraints is for us to make sure that we are talking about the same triangle which should be the smallest triangle among the many triangles created from the great circles (or at least those without a segment of a great circle in it).

5 Hyperbolic Geometry

5.1 Lorentz Products

We first develop the necessary tools to develop the hyperbolic space.

Definition 5.1.1 (Lorentz Inner Product) Let $x, y \in \mathbb{R}^n$. Define the Lorentz inner product to be

$$\langle x, y\rangle_L = -x_1y_1 + x_2y_2 + x_3y_3 + \cdots + x_ny_n$$

Definition 5.1.2 (Lorentz Norm) Define the Lorentz norm of a vector $x \in \mathbb{R}^n$ to be

$$\|x\|_L = \sqrt{\langle x, x\rangle_L}$$

where ther square root to be positive or positive imaginary or 0.

Note that the Lorentz norm allows imnaginary numbers and it could also be negative and thus is not a norm in the usual sense.

Proposition 5.1.3 (Hyperbolic Polarization Identity) Let $x, y \in \mathbb{R}^n$. Then

$$\langle x, y\rangle_L = \frac{1}{4}\|x+y\|_L^2 - \frac{1}{4}\|x-y\|_L^2$$

Definition 5.1.4 (Classification of Vectors) Let $x \in \mathbb{R}^n$. We say that x is
- space-like if $\|x\|_L > 0$
- light-like if $\|x\|_L = 0$
- time-like if $\|x\|_L \in \mathbb{C} \setminus \mathbb{R}$

Definition 5.1.5 (Lorentz Orthnormal) A set of vectors $v_1, \dots, v_n \in \mathbb{R}^n$ is Lorentz Orthgonal if $\langle v_i, v_j\rangle_L = 0$ for $i \neq j$. They are said to be Lorentz Orthonormal if

$$\langle v_i, v_j\rangle_L = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } 2 \leq i = j \leq n \\ -1 & \text{if } i = j = 1 \end{cases}$$

Lemma 5.1.6 If $v_1, \dots, v_n$ are Lorentz Orthonormal, then they form a basis of $\mathbb{R}^n$.

Proof We just have to show that they are linearly independent. Suppose that $\sum_{k=1}^n a_kv_k = 0$ for some $a_1, \dots, a_n \in \mathbb{R}$. For each $i \in \{2, \dots, n\}$, we have that

$$0 = \left\langle \sum_{k=1}^n a_kv_k, v_i \right\rangle_L = a_i$$

and for $i = 1$ we have that the inner product is $-a_1$. Thus $a_1 = \cdots = a_n = 0$ and we are done. ■

Definition 5.1.7 (Lorentz Cross Product) Let $x, y \in \mathbb{R}^3$. Define the Lorentz cross product of x and y to be

$$x \times_L y = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} (x \times y) = \begin{vmatrix} -e_1 & e_2 & e_3 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix}$$

An easy way to remember everything for Lorentz related operations is that the first element in a coordinate related operation is always negative.

Lemma 5.1.8 (Hyperbolic Binet Cauchy Identity) For $x, y, z, w \in \mathbb{R}^3$, the following identity holds.
$$\langle x \times_L y, z \times_L w \rangle_L = -\langle x, z \rangle_L \cdot \langle y, w \rangle_L + \langle x, w \rangle_L \cdot \langle y, z \rangle_L$$

Now that we have mostly developed a sufficient amount linear algebra, we turn to study matrices that corresponds to Lorentz operations.

Definition 5.1.9 (Lorentz Orthgonal Matrices) Let $J = \begin{pmatrix} -1 & 0 \\ 0 & I_{n-1} \end{pmatrix}$. A $n \times n$ real matrix is Loretnz orthgonal if
$$A^T J A = J$$

Proposition 5.1.10 The following are true for any $x, y \in \mathbb{R}^n$.
- $\langle x, y \rangle_L = x^T J y = \langle Jx, y \rangle$
- $x \times_L y = J(x \times y) = (Jy) \times (Jx)$

J will often replace the function of the identity in Hyperbolic space. From time to time we will see it reappear.

Proposition 5.1.11 Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a linear map. The following are equivalent.
- T is a Lorentz Transformation
- The matrix of T is Lorentz Orthogonal
- $\|T(x)\|_L = \|x\|_L$ for all $x \in \mathbb{R}^n$

Proof

- $(1) \iff (2)$: Suppose that T is a Lorentz transformation. Let the matrix A represent T inthe standard basis so that it has columns $a_1, \dots, a_n$. This means that $T(e_k) = a_k$. Then we have
$$\begin{aligned} A^T J A &= \begin{pmatrix} a_1^T \\ \vdots \\ a_n^T \end{pmatrix} J \begin{pmatrix} a_1 & \cdots & a_n \end{pmatrix} \\ &= \begin{pmatrix} a_1^T \\ \vdots \\ a_n^T \end{pmatrix} \begin{pmatrix} Ja_1 & \cdots & Ja_n \end{pmatrix} \\ &= \begin{pmatrix} a_1^T Ja_1 & \cdots & a_1^T Ja_n \\ \vdots & \ddots & \vdots \\ a_n^T Ja_1 & \cdots & a_n^T Ja_n \end{pmatrix} \\ &= \begin{pmatrix} \langle a_1, a_1 \rangle_L & \cdots & \langle a_1, a_n \rangle_L \\ \vdots & \ddots & \vdots \\ \langle a_n, a_1 \rangle_L & \cdots & \langle a_n, a_n \rangle_L \end{pmatrix} \end{aligned}$$
Thus we can see that $A^T J A = J$ if and only if $T(e_1), \dots, T(e_n)$ are Lorentz orthonormal if and only if T is a Lorentz transformation.
- $(1) \implies (3)$: Suppose that T is a Lorentz transoformation. Then trivially
$$\|T(x)\|_L^2 = \langle T(x), T(x) \rangle_L = \langle x, x \rangle_L = \|x\|_L^2$$
Taking the positive square root gives our result.
- $(3) \implies (1)$: Using the polarization identity, we have that

■

5.2 Lorentz Transformations and $O(1, n)$

Definition 5.2.1 (Lorentz Transformation) A map $\phi : \mathbb{R}^n \to \mathbb{R}^n$ is a Lorentz transformation if it preserves the Lorentz inner product:

$$\langle \phi(x), \phi(y) \rangle_L = \langle x, y \rangle_L$$

for all $x, y \in \mathbb{R}^n$. We say that ϕ is positive if both x and $\phi(x)$ has their first coordinate larger than 0.

Proposition 5.2.2 A map $T : \mathbb{R}^n \to \mathbb{R}^n$ is a Lorentz Transformation if and only if $\{T(e_1), \dots, T(e_n)\}$ is a Lorentz orthonormal basis and T is linear.

Proof Suppose that T is a Lorentz transformation. Then since it preserves the Lorentz inner product and $e_1, \dots, e_n$ is trivially Lorentz orthonormal, we know that $T(e_1), \dots, T(e_n)$ are Lorentz orthonormal and we are done. Now let $x = \sum_{k=1}^n x_k e_k$ and $T(x) = \sum_{k=1}^n b_k T(e_k)$. Then

$$-b_1 = \langle T(x), T(e_1) \rangle_L = \langle x, e_1 \rangle_L = -x_1$$

and

$$b_k = \langle T(x), T(e_k) \rangle_L = \langle x, e_k \rangle_L = -x_k$$

for $k \in \{2, \dots, n\}$ thus T is linear.
Suppose now that $T(e_1), \dots, T(e_n)$ is Lorentz orthonormal and T is linear. Then

$$\begin{aligned}
\langle T(x), T(y) \rangle_L &= T(x)^T J T(y) \\
&= \left(\sum_{k=1}^n x_k T(e_k) \right)^T J \left(\sum_{k=1}^n y_k T(e_k) \right) \\
&= \sum_{k=1}^n \sum_{j=1}^n x_k y_j T(e_k)^T J T(e_j) \\
&= \sum_{k=1}^n \sum_{j=1}^n x_k y_j e_k J e_j \\
&= \left(\sum_{k=1}^n x_k e_k \right)^T J \left(\sum_{k=1}^n y_k e_k \right) \\
&= \langle x, y \rangle_L
\end{aligned}$$

■

Definition 5.2.3 (Lorentz Group) The Lorentz group is the group of Lorentz orthogonal $n + 1, n + 1$ matrices denoted

$$O(1, n) = \{A \in M_{n+1\times n+1}(\mathbb{R})\} | A^T J A = J\}$$

Proposition 5.2.4 $O(1, n)$ is a group.

Proof ■

Proposition 5.2.5 Let $A \in M_{m\times n}(\mathbb{R})$. Then the following are equivalent.

- The map $T(x) = Ax$ is a Lorentz transformation
- $\|T(x)\|_L = \|x\|_L$ for all $x \in \mathbb{R}^n$
- A is Lorentz orthogonal.

Corollary 5.2.6 Every Lorentz Transformation corresponds to an element of $O(1,n)$ and every element of $O(1,n)$ gives rise to a Lorentz Transformation.

Proof Let T be a lorentz transformation.
Now let T be a hyperbolic isometry. Define a Lorentz transformation S by $S|_{H^n} = T$ ■

5.3 Positive Lorentz Transformations

Definition 5.3.1 (Lorentz Transformation) A map $\phi : \mathbb{R}^n \to \mathbb{R}^n$ is a positive Lorentz transformation if the following are true

- ϕ is a Lorentz transformation
- x is time-like if and only if $T(x)$ is time-like for $x \in \mathbb{R}^n$.

Definition 5.3.2 (Positive Lorentz Group) The positive Lorentz group is the set of all elements in $O(1,n)$ which maps positive time-like vectors bijectively to positive time-like vectors, denoted $O^+(1,n)$

Proposition 5.3.3 $O^+(1,n)$ is a group.

Proof ■

Lemma 5.3.4 Let $A \in O(1,n)$. Then $A \in O^+(1,n)$ if and only if $a_{1,1} > 0$.

This is extremely powerful since it gives a fairly easy way to check whether a Lorentz orthogonal matrix is an isometry.

Corollary 5.3.5 Every Positive Lorentz Transformation corresponds to an element of $O^+(1,n)$ and every element of $O^+(1,n)$ gives rise to a Positive Lorentz Transformation.

Proof Let T be a lorentz transformation.
Now let T be a hyperbolic isometry. Define a Lorentz transformation S by $S|_{H^n} = T$ ■

Lemma 5.3.6 For every set of linearly independent $a, b, c \in H^n$, there exists an element $A \in O^+(1,n)$ such that

$$\begin{aligned} Aa &= (1, 0, \dots, 0) \\ Ab &= (b_1, b_2, 0, \dots, 0) \text{ with } b_1 > 0 \\ Ac &= (c_1, c_2, c_3, 0 \dots, 0) \text{ with } c_1 > 0 \end{aligned}$$

Proof Let $a, b, c, z_4, \dots, z_{n+1}$ be a basis for H^n. We construct a Lorentz orthonormal basis for $\mathbb{R}^{n+1}$. Since $a \in H^n$ implies $\|a\| = i$, we can choose $v_1 = a$. We find the basis by induction, where v_k is constructed by $a, b, c, z_4, \dots, z_k$. Now let $w_2 = b + \langle v_1, b\rangle_L v_1$. Observe that

$$\begin{aligned} \langle w_2, v_1\rangle_L &= \langle b, v_1\rangle_L + \langle v_1, b\rangle_L \langle v_1, v_1\rangle_L \\ &= \langle b, v_1\rangle_L - \langle v_1, b\rangle_L && (\langle v_1, v_1\rangle_L = -1) \\ &= 0 \end{aligned}$$

Thus w_2 is Lorentz orthogonal to v_1. Now just choose $v_2 = \frac{w_2}{\|w_2\|_L}$. Note that this is similar to the Gram-schimdt procedure in Euclidean Space. In general, suppose that $v_1, \dots, v_k$ are now made orthogonal. Define

$$w_{k+1} = a - \sum_{i=1}^{k} \langle z_{k+1}, v_i\rangle_L v_i$$

Then w_{k+1} will be orthogonal to all of $v_1, \dots, v_k$ thus we can choose $v_{k+1} = \frac{w_{k+1}}{\|w_{k+1}\|}$. Thus a Lorentz orthonormal basis $v_1, \dots, v_{n+1}$ is formed.

With respect to this basis, we have the desired results. ■

5.4 Hyperbolic Space

Definition 5.4.1 (Hyperbolic Space) The n dimensional hyperbolic space is the space

$$H^n = \{x \in \mathbb{R}^{n+1} | \|x\|_L = i \text{ and } x_1 > 0\}$$

along with the metric hyperbolic metric

$$d_{H^n}(x, y) = \cosh^{-1}(-\langle x, y\rangle_L)$$

In order to prove that H^n is a metric space, we need to first develop the notion of lines and triangles and angles in H^n.

Definition 5.4.2 (Classification of Subspaces) We say that a vector subspace of $\mathbb{R}^n$ is

- time-like if V has at least one time-like vector
- space-like if every nonzero vector in V is space-like
- light-like otherwise

To check the type of subspace, we first find whether there is one time-like vector, we then check whether there is one light-like vector.

Proposition 5.4.3 There is a parametrization between $\mathbb{R}^2$ and H^2 given by

$$f(t, \theta) = (\cosh(t), \cos(\theta)\sinh(t), \sin(\theta)\sinh(t))$$

for $t \in [0, \infty)$ and $\theta \in [0, 2\pi]$ where $\mathbb{R}^2$ is in polar coordinates and H^2 in cartesian coordinates.

Proof ■

Definition 5.4.4 (Hyperbolic Lines and Lorentz Planes) A Lorentz plane is a 2 dimensional vector subspace of $\mathbb{R}^{n+1}$ that contains a timelike vector. A hyperbolic line is the intersection of H^n with any Lorentz Plane.

Definition 5.4.5 (Collinearity) Three points $x, y, z \in H^n$ are hyperbolically collinear if and only if there is a hyperbolic line L of H^n with $x, y, z \in L$.

We need the following theorem to show that d_{H^n} is a metric.

Lemma 5.4.6 For every $P \neq Q \in H^n$, there exists a unique hyperbolic line L containing P and Q.

Lemma 5.4.7 Every line through the origin in $\mathbb{R}^{n+1}$ intersects H^n in at most 1 point.

Proof Let $y = \lambda x$ for some $x, y \in H^n$. Then by definition of H^n, we must have

$$\langle x, x\rangle_L = \langle y, y\rangle_L = -1$$

But the Lorentz inner product is linear. Thus $\langle y, y\rangle_L = \lambda^2\langle x, x\rangle_L$ and thus $\lambda = \pm 1$. Since both x_1 and y_1, the first component of x and y must be positive, we must have $\lambda = 1$ and we are done. ■

Definition 5.4.8 (Hyperbolic Triangles) A hyperbolic triangle, denoted $\triangle PQR$, consists of three distinct, non-collinear points $P, Q, R \in H^n$, and the finite hyperbolic line segments joining each pair of points, and the finite area enclosed by these lines.

Definition 5.4.9 (Hyperbolic Angles) Let $\triangle PQR$ be a hyperbolic triangle in $H^2 \subset \mathbb{R}^3$. Define the hyperbolic angle a at P to be $a \in [0, \pi]$ such that

$$\cos(a) = \frac{\langle P \times_L R, P \times_L Q \rangle_L}{\|P \times_L R\|_L \|P \times_L Q\|_L}$$

To prove the triangle inqeuality for d_{H^n} we need the following theorem.

Theorem 5.4.10 Let $x, y, z \in H^n$ be distinct. Let $\alpha = d_{H^n}(z, y)$, $\beta = d_{H^n}(x, z)$, $\gamma = d_{H^n}(x, y)$ and a be the hyperbolic angle of $\triangle xyz$ at x. Then

$$\cosh(\alpha) = \cosh(\beta) \cdot \cosh(\gamma) - \sinh(\beta) \cdot \sinh(\gamma) \cdot \cos(a)$$

Proof WLOG we can assume x, y, z takes the form $x = (1, 0, \dots, 0)$, $y = (\cosh(t_1), \sinh(t_1), 0, \dots, 0)$ and $z = (\cosh(t_2), \cos(\theta)\sinh(t_2), \sin(\theta)\sinh(t_2), 0, \dots, 0)$. This is because Loretnz transformation preserves the Loretnz inner product and thus the distance function remains unchanged. Since the trailing zeroes make no contributions to the calculations, we assume we are working on H^2. Now note that

$$x \times_L \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} 0 \\ -a_3 \\ a_2 \end{pmatrix}$$

Thus we have

$$\begin{aligned} \cos(a) &= \frac{\langle x \times_L y, x \times_L z \rangle_L}{\|x \times_L y\|_L \|x \times_L z\|_L} \\ &= \frac{1}{|\sinh(t_1)||\sinh(t_2)|} \left\langle \begin{pmatrix} 0 \\ 0 \\ \sinh(t_1) \end{pmatrix}, \begin{pmatrix} 0 \\ -\sin(\theta)\sinh(t_2) \\ \cos(\theta)\sinh(t_2) \end{pmatrix} \right\rangle_L \\ &= \cos(\theta) \end{aligned}$$

Now we have that

$$\begin{aligned} \cosh(\gamma) &= -\langle x, y \rangle_L = \cosh(t_1) \\ \cosh(\beta) &= -\langle x, z \rangle_L = \cosh(t_2) \\ \cosh(\alpha) &= -\langle z, y \rangle_L = \cosh(t_1)\cosh(t_2) - \cos(\theta)\sinh(t_1)\sinh(t_2) \end{aligned}$$

Combining these and using the fact that $\cos(a) = \cos(\theta)$ gives our result. ∎

Theorem 5.4.11 The hyperbolic metric is indeed a metric on H^n and thus the hyperbolic space is a metric space.

Proof

- We first prove that the hyperbolic metric is well defined. This means that we need to prove that $-\langle x, y \rangle_L \geq 1$. Let $x, y \in H^n$. Denote $x' = (1, x_2, \dots, x_{n+1})$ and $y' = (1, y_2, \dots, y_{n+1})$. Then we have that
$$x_1^2 = 1 + x_2^2 + \dots + x_n^2 = \|x'\|^2$$
and likewise for y_1^2. Note that the norm on the right is the usual Euclidean norm. By Cauchy

Schwarz we have that $\langle x', y'\rangle^2 \leq \|x'\|^2\|y'\|^2 = (x_1y_1)^2$ which implies that

$$|1 + x_2y_2 + \cdots + x_ny_n| \leq |x_1y_1| = x_1y_1$$

Rearraging, we have that $\langle x, y\rangle_L \leq -1$ and thus

$$-\langle x, y\rangle_L \geq 1$$

- We want to show that $d_{H^n}(x, y) \geq 0$ with equality if and only if $x = y$. Trivially the image of $\cosh^{-1}$ is greater than or equal to 0 and $\cosh^{-1}(u) = 0$ means that $u = 1$ and $-\langle x, y\rangle_L = 1$. From the above proof, in order for this to be true, we need $\langle x', y'\rangle^2 = \|x'\|^2\|y'\|^2$ which is true if and only if x is a multiple of y. But we know that every line passing through the origin only intersects H^n in at most one point. Thus $x = y$ and we are done.
- We want to showt that $d_{H^n}(x, y) = d_{H^n}(y, x)$ for all $x, y \in H^n$. But clearly the Lorentz inner product is symmetric, thus we are done.
- To show the triangle inequality, let $x, y, z \in H^n$. Let $\alpha = d_{H^n}(z, y)$, $\beta = d_{H^n}(x, z)$, $\gamma = d_{H^n}(x, y)$ and a be the hyperbolic angle of $\triangle xyz$ at x. Using the above theorem, we find that

$$\cosh(\alpha) \leq \cosh(\gamma)\cosh(\beta) + \sinh(\beta)\sinh(\gamma) = \cosh(\beta + \gamma)$$

Since $\cosh(x)$ is increasing on $[0, \infty)$, this implies that $\alpha \leq \beta + \gamma$. Thus we are done.

■

Proposition 5.4.12 Let $R \in \text{Isom}(H^n, d_H)$. Then there exists a unique $A \in O^+(1, n)$ with $R = T_A|_{H^n}$, where T_A is the linear map on $\mathbb{R}^{n+1}$ given by $T(x) = Ax$.

Corollary 5.4.13 There is a group isomorphism between $\text{Isom}(H^n)$ and $O^+(1, n)$.

Becareful that all isometries of H^n is not the Loretnz group but only of the positive Lorentz group.

5.5 Hyperbolic Lines

Lemma 5.5.1 If $x \in \mathbb{R}^n$ with $\langle x, x\rangle_L < 0$, and $w \in \mathbb{R}^n \setminus \{0\}$ with $\langle x, w\rangle_L = 0$, then $\langle w, w\rangle_L > 0$. In other words, any vector that is Lorentz orthogonal with a time-like vector must be a space-like vector.

Proof Let $x' = x - x_1e_1$ and $w' = w - w_1e_1$. Then $\langle x', x'\rangle_L \geq 0$ and $\langle x, x\rangle_L = -x_1^2 + \langle x', x'\rangle_L$. This means that $x_1 \neq 0$. If $w_1 = 0$, then since $w \neq 0$, we must have $\langle w, w\rangle_L = \langle w', w'\rangle_L > 0$ and we are done.
Now suppose that $w_1 \neq 0$. Then $x_1w - w_1x$ has zero e_1 component. I claim that this vector is non-zero. Indeed if $x_1w = w_1x$, then

$$0 = \frac{x_1}{w_1}\langle x, w\rangle_L = \langle x, \frac{x_1}{w_1}w\rangle_L = \langle x, x\rangle_L < 0$$

Now we have that

$$\begin{aligned} 0 \leq \|x_1w - w_1x\|_L^2 &= \langle x_1w - w_1x, x_1w - w_1x\rangle_L \\ &= x_1^2\langle w, w\rangle_L + w_1^2\langle x, x\rangle_L \end{aligned}$$

Bu this means that

$$\langle w, w\rangle_L \geq -\frac{w_1^2}{x_1^2}\langle x, x\rangle_L > 0$$

and we are done. ■

In layman terms, this means that the orthogonal subspace of a time-like vector is a subspace with all vectors being space-like.

Definition 5.5.2 Let L_1, L_2 be two distinct lines in H^2 with Lorentz plane Π_1 and Π_2 respectively. Let $v \in \Pi_1 \cap \Pi_2 \setminus \{0\}$. Let $V = \Pi_1 \cap \Pi_2$.
- If V is time like, L_1 and L_2 intersect at a point x and $V = \mathbb{R}v = \Pi_1 \cap \Pi_2$
- If V is space like, L_1 and L_2 are parallel and diverge
- If V is light like, L_1 and L_2 are ultraparallel

The following theorem shows that Euclid's Postulate does not hold in hyperbolic space.

Theorem 5.5.3 Let $x \in H^2$. Let L be a line in H^2. Then there are infinitely many lines in H^2 which pass through x and do not intersect L.

5.6 Hyperbolic Triangles and Angles

Proposition 5.6.1 The hyperbolic angles is invariant under hyperbolic isometries.

Proposition 5.6.2 Let $\triangle PQR$ be a hyperbolic triangle. Let a, b, c be the angles at P, Q, R respectively. Then

$$a + b + c = \pi - \text{Area}(\triangle PQR)$$

5.7 Normal Form Theorem

Lecture did not have time to thoroughly prove things and so do I. Every Lorentz Transformation must have one of the following four forms:
Lorentz Translation:

$$\begin{pmatrix} \cosh(\beta) & \sinh(\beta) & 0 \\ \sinh(\beta) & \cosh(\beta) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

This transformation does not have a fixed point in H^2. Orientation is preserved.
Lorentz Glide:

$$\begin{pmatrix} \cosh(\beta) & \sinh(\beta) & 0 \\ \sinh(\beta) & \cosh(\beta) & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

This transformation does not have a fixed point in H^2. Orientation is also reversed
Rotation about the point $(1, 0, 0)$:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cosh(\beta) & \sinh(\beta) \\ 0 & \sinh(\beta) & \cosh(\beta) \end{pmatrix}$$

This transformation has one fixed point, namely $(1, 0, 0)$ and is orientation preserving.
Reflection along $x_2 = 0$:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

This transformation fixes the line $x_2 = 0$ and is orientation reversing.

6 Projective Geometry

6.1 Projective Space

Lemma 6.1.1 Let $\mathbb{F}$ be a field. The relation $\sim$ in $\mathbb{F}^{n+1}$ where $(x_0, \dots, x_n) \sim (y_0, \dots, y_n)$ if and only if $y_i = \lambda x_i$ for all $i \in \{1, \dots, n\}$ with $\lambda \in \mathbb{F}$ is an equivalence relation.

Definition 6.1.2 (Projective Space) The equivalence relation $\sim$ on $\mathbb{F}^{n+1}$ induces the projective space with elements in it being 1 dimensional subspaces of $\mathbb{F}^n$, written as

$$\mathbb{P}(\mathbb{F}^{n+1}) = \frac{\mathbb{F}^{n+1} \setminus \{0\}}{\sim}$$

We use $\mathbb{P}^n$ to denote $\mathbb{P}(\mathbb{R}^{n+1})$. Also define the dimension of $\mathbb{P}^n$ to be n.

We use $\mathbb{R}^{n+1}$ as our vector space since every finite dimensional vector space is isomorphic to $\mathbb{R}^n$ for some n.

Proposition 6.1.3 There is a bijection between $\mathbb{P}^n$ and the set of lines through the origin in $\mathbb{R}^{n+1}$.

Proposition 6.1.4 We have that

$$\mathbb{P}^n \cong \mathbb{R}^n \cup \mathbb{R}^{n-1} \cup \cdots \cup \mathbb{R} \cup \{\infty\}$$

Proof The proof is done by induction. We first consider $\mathbb{P}(\mathbb{R})$. Recall that $x \sim y$ if and only if $y = tx$ for some $t \neq 0$. This means that $[x]$ represents the same point as long as $x \in \mathbb{R} \setminus \{0\}$. We also simply identify $[0]$ with ∞.
We also consider $\mathbb{P}(\mathbb{R}^2)$ for better illustration. All elements of $\mathbb{P}(\mathbb{R}^2)$ are of the form $[x_0 : x_1]$. We consider the case that $x_0 \neq 0$. Since $x_0 \neq 0$, we can divide out x_0 to get the same coordinate $[1 : x_1] \in \mathbb{P}(\mathbb{R}^2)$. For each $x_1 \in \mathbb{R}$, $[1 : x_1]$ represents different coordinates in $\mathbb{P}(\mathbb{R}^2)$ thus we can easily identify it with $\mathbb{R}$. Now if $x_0 = 0$, $[0 : x_1]$ would represent the same coordinate for any $x_1 \in \mathbb{R}$. This is precisely the same situation as $\mathbb{P}(\mathbb{R})$. Thus we can virtually ignore the first coordinate and identify it with $\mathbb{P}(\mathbb{R})$. Thus we have shown that $\mathbb{P}(\mathbb{R}^2) = \mathbb{R}^2 \cup \mathbb{P}(\mathbb{R})$.
Now through the induction hypothesis, we just have to show that $\mathbb{P}(\mathbb{R}^n) = \mathbb{R}^n \cup \mathbb{P}(\mathbb{R}^{n-1})$. But this is done in a similar fashion. We can use $[1 : x_1 : \cdots : x_n]$ for $x_1, \dots, x_n \in \mathbb{R}$ to identify $\mathbb{R}^n$ and $\mathbb{P}(\mathbb{R}^{n-1})$ with $[0 : x_1 : \cdots : x_n]$. ∎

Theorem 6.1.5 We have that

$$\mathbb{P}(\mathbb{R}^n) = \frac{S^n}{\pm}$$

where $\pm$ is the equivalence relation of antipodal points.

6.2 Projective Linear Subspaces

Definition 6.2.1 (Projective Linear Subspaces) Let $U \subseteq V$ be a vector subspace. Define the projective linear subspace to be

$$\mathbb{P}(U) = \frac{U \setminus \{0\}}{\sim} \subseteq \mathbb{P}(V)$$

with its dimension defined to be $\dim(\mathbb{P}(U)) = \dim(U) - 1$. We define $\dim(\emptyset) = -1$ for conventions.

Definition 6.2.2 (Classification of Subspaces) Let $U \subseteq V$ be a vector subspace where $\dim(V) = n$. We say that $\mathbb{P}(U)$ is

- A single point if $\dim(U) = 1$, or $\dim(\mathbb{P}(U)) = 0$
- A line if $\dim(U) = 2$, or $\dim(\mathbb{P}(U)) = 1$
- A plane if $\dim(U) = 3$, or $\dim(\mathbb{P}(U)) = 2$

- A hyperplane if $\dim(U) = n - 1$, or $\dim(\mathbb{P}(U)) = n - 2$

In particular, note that lines in $\mathbb{R}^{n+1}$ become points. Let $x \in \mathbb{R}^{n+1}$ be fixed. Take the line $y = tx$ for $t \in \mathbb{R}$ as an example. Observe that in the projective space. $x \sim y$ if and only if $y = tx$ for some $t \in \mathbb{R}$. Naturally $y = tx$ becomes a point.

Definition 6.2.3 (Projective Cone and Span) Let V be a vector subspace and let $U \subset \mathbb{P}(V)$ be a subset. Define the projective cone of U to be

$$\tilde{U} = \bigcup_{v \in U} v$$

Define the span of U to be

$$\langle U \rangle = \mathbb{P}(\text{span}(\tilde{U}))$$

the smallest projective linear subspace of $\mathbb{P}(V)$ containing U.

Theorem 6.2.4 (Dimemsion Theorem) Let $E, F \subset \mathbb{P}^n$ be projective linear subspaces. Then

$$\dim(E \cap F) = \dim(E) + \dim(F) - \dim\langle E, F \rangle$$

Proof Let ∎

Theorem 6.2.5 Any two distinct lines L_1 and L_2 in $\mathbb{P}^2$ intersect in a point.

Proof Note that L_1 and L_2 are projections of two planes A_1, A_2 through the origin. Since A_1, A_2 have dimension 2 and are distinct. Thus their intersection must have dimension 1 and their span has dimension 3. But $\langle L_1, L_2 \rangle = \mathbb{P}(\text{span}(A_1, A_2)) = \mathbb{P}(\mathbb{R}^3)$ and thus $\dim(\langle L_1, L_2 \rangle) = 2$. By the above theorem, we must have $\dim(L_1 \cap L_2) = 0$ and thus it is exactly a point. ∎

6.3 Projective Transformations

Definition 6.3.1 (Projective Transformations) A projective transformation of $\mathbb{P}^n$ is a map $T|_A : \mathbb{P}^n \to \mathbb{P}^n$ defined by

$$T([x]) = [Ax]$$

where $A \in \text{GL}(n+1, \mathbb{R})$.

Definition 6.3.2 (Projective General Linear Group) The projective general linear group of a vector space V over k with dimension n is the group of all invertible linear transformations unique up to scalar multiplication. That is, we say that $T_1 \sim T_2$ if $T_1 = \lambda T_2$ for some $\lambda \in k$ and $\lambda \neq 0$. It is denoted as $\text{PGL}(n+1, k)$.

Definition 6.3.3 (Projective Frame of Reference) A projective frame of reference for $\mathbb{P}^n$ is an ordered set of $n+2$ points, $P_0, \dots, P_{n+1} \in \mathbb{P}^n$ such that any $n+1$ points are linearly independent. The standard frame of reference is just $[e_1], \dots, [e_{n+1}], [e_1 + \cdots + e_{n+1}]$.

Proposition 6.3.4 There is a bijection between projective transformations of $\mathbb{P}^n$ and projective frames of references of $\mathbb{P}^n$ as follows:

$$\phi(T) = \{T([e_1]), \dots, T([e_{n+1}]), T([e_1 + \cdots + e_{n+1}])\}$$

where T is the projective transformation.

This means that specifying any $n+2$ points in $\mathbb{P}^n$ such that they form a projective frame of reference induces a unique projective transformation.

6.4 Perspectivities

Definition 6.4.1 (Perspectivities) Let Π_1, Π_2 be two hyperplanes in $\mathbb{P}^n$. A perspectivity $f : \Pi_1 \to \Pi_2$ from a point $O \in \mathbb{P}^n$ but not in Π_1 or Π_2 is a map given by

$$f(P) = \Pi_2 \cap \langle O, P \rangle$$

Lemma 6.4.2 Perspectivities are well defined.

Proof We want to show that $f(P)$ is a point. We appeal to the dimension theorem. We have that

$$\dim(\Pi_2 \cap \langle O, P \rangle) = \dim(O) + \dim(\Pi_2) - \dim(\langle \Pi_2 \langle O, P \rangle \rangle)$$

Since O is necessarily not in Π_1, we have $\langle O, P \rangle$ is a projective line. Moreover, since O is not in Π_2, we must have that the span of the hyperplane Π_2 and a line $\langle O, P \rangle$ must be the projective space itself. Thus we have that

$$\dim(\Pi_2 \cap \langle O, P \rangle) = 0$$

This means that $f(P)$ must be a point in projective space. ■

Definition 6.4.3 (Cross Ratio) Let P, Q, R, S be distinct and ordered points on a projective line in $\mathbb{P}^n$. Define the cross ratio between the four projective points to be the ratio

$$(P, Q; R, S) = \left(\frac{p-r}{p-s}\right)\left(\frac{q-s}{q-r}\right)$$

where the ratio between vectors lying on the same projective line is defined to be the λ in $\frac{\lambda v}{v}$.

Proposition 6.4.4 Projective linear maps perserve perspectivity.

Proof By linearity, we must have

$$\frac{T(\lambda v)}{T(v)} = \frac{\lambda v}{v}$$

■

Lemma 6.4.5 Perspectivities are projective linear maps.

6.5 Important Theorems

Definition 6.5.1 (Triangles in Projective Space) Let $P, Q, R \in \mathbb{P}^n$ be distinct. The triangle $\triangle PQR$ in $\mathbb{P}^n$ is defined to be the three points and the three sides spanned by three pair of points.

Definition 6.5.2 (In Perspective from a Point) Two triangles $\triangle PQR$ and $\triangle P'Q'R'$ in $\mathbb{P}^n$ are said to be in persepective from a point O if the lines passing through PP', QQ' and RR' intersect at a O.

Definition 6.5.3 (In Perspective from a Line) Two triangles $\triangle PQR$ and $\triangle P'Q'R'$ in $\mathbb{P}^n$ are said to be in persepective from a line L if the points $PQ \cap P'Q'$, $QR \cap Q'R'$ and $PR \cap P'R'$ are collinear.

Note that this definition makes sense since any two lines in projective space must meet at one point.

Theorem 6.5.4 (Desargues' Theorem) If $\triangle PQR$ and $\triangle P'Q'R'$ are two distinct triangles in $\mathbb{P}^n$ which are in perspective from a point, then they are also in perspective from a line.

Theorem 6.5.5 (Pappus' Theorem) Let L and L' be distinct projective lines in $\mathbb{P}^2$. Let P, Q, R in L but not in L' and P', Q', R' in L' but not in L all be distinct points. Then the intersection points $PQ' \cap P'Q$, $PR' \cap P'R$ and $QR' \cap Q'R$ are collinear.

6.6 Axiomatic Projective Geometry

Just as Euclidean geometry has 5 axioms to follow, we can also establish axioms for projective geometry.

Definition 6.6.1 (Axiomatic Projective Geometry) An axiomatic projective plane (P, L, I) consists of a set P called the set of points, a set L, the set of lines and a relation $I \subseteq P \times L$, the incidence relation. For $l_1, l_2 \in L$, define

$$l_1 \cap l_2 = \{p \in P | p \in l_1, p \in l_2\}$$

These sets must satisfy the following four axioms:

1. Every line contains at least three distinct points:
$$\forall l \in L \exists \text{ distinct } x, y, z \in P \text{ such that } (x, l) \in I, (y, l) \in I, (z, l) \in I$$
2. Every point is contained in at least three distinct lines:
$$\forall x \in P \exists \text{ distinct } l, m, n \in L \text{ such that } (x, l) \in I, (x, m) \in I, (x, n) \in I$$
3. Any two points span a unique line:
$$\forall x \neq y \in P \exists ! l \in L \text{ such that } (x, l) \in I, (y, l) \in I$$
4. Any two distinct lines intesect at a unique point:
$$\forall l \neq m \in I \exists ! x \in P \text{ such that } (x, l) \in I, (x, m) \in I$$